



Figure 3: Representation-quality metrics for human and machine latent representations from the perspectives of information content (Entropy, Effective Rank) and geometric structure (Anisotropy, Intrinsic Dimensionality). **Takeaway:** Machine-text representations are more compressed, concentrated along fewer dominant directions, and located on simpler manifolds than human-written text representations. These structural differences help explain why the two are linearly separable in latent space.

representations from the perspective of *information content* (Entropy, Effective Rank) and *geometric structure* (Anisotropy, Intrinsic Dimension) representation quality metrics (Skean et al., 2025): (1) **Entropy** (Sanchez Giraldo et al., 2015) measures how much information a representation contains; (2) **Effective Rank** (Roy and Vetterli, 2007) quantifies the amount of non-redundant information in a representations; (3) **Anisotropy** (Razhigaev et al., 2024) measures the degree to which activations concentrate along specific directions in latent space; and (4) **Intrinsic Dimension** (Facco et al., 2017) estimates the minimum number of dimensions required to describe the geometric structure of representations. Appendix A.1 provides additional intuition and formulae.

MGT representations are more compressed, while HWT representations contain richer and more diverse features. Figure 3 plots representation-quality metrics across layers for HWT and MGT activations. The information-theoretic metrics (Entropy and Effective Rank) capture how broadly information is distributed across representations. Higher values indicate that variance is spread across many dimensions, whereas lower values indicate that information collapses into fewer dominant components. Across layers, HWT exhibits substantially higher entropy than MGT, suggesting that human-written text activates a broader and less redundant set of features. In contrast, MGT has much lower entropy, indicating a more compressed representation in which activations carry less diverse information. With the same pattern appearing for Effective Rank, both metrics suggest that MGT representations occupy a more information-compressed subspace, while HWT retains richer, more heterogeneous latent features.

MGT occupies a narrower and lower-dimensional activation geometry than HWT.

The geometric metrics (Anisotropy and Intrinsic Dimension) characterize the shape of activation distributions. Higher anisotropy indicates that representations align strongly with a small number of dominant directions, whereas higher intrinsic dimensionality indicates that more dimensions are required to capture their local structure. Across most middle and later layers, MGT activations are more anisotropic than HWT activations, suggesting that MGT concentrates along a narrower set of directions in latent space. At the same time, MGT exhibits consistently lower intrinsic dimensionality, indicating that its representations lie on a simpler, lower-dimensional manifold. HWT shows the opposite pattern: lower anisotropy and higher intrinsic dimensionality suggest that its activations are more uniformly distributed and occupy a more complex manifold. This finding is consistent with prior work reporting lower intrinsic dimensionality for MGT (Tulchinskii et al., 2023). We further show that this structure emerges systematically across layers and remains consistent across diverse domains.

Takeaway Our linearity and representation analyses support three takeaways: (1) HWT and MGT representations are linearly separable in low-dimensional latent space beginning early layers; (2) information-content metrics show that MGT representations are more compressed, whereas HWT representations are richer and more diverse; and (3) geometric metrics show that MGT representations occupy a narrower, more anisotropic, and lower-dimensional manifold, whereas HWT spans a broader and more complex region of activation space. These structural differences provide a plausible explanation of *why* simple linear MGT detectors likely are sufficient and effective in latent space.

4 Methodology

The linear separation between MGT and HWT representations, driven by systematic distributional differences in latent space, motivates the use of **MGT probes**. Drawing on the interpretability literature (Alain and Bengio, 2016; Belinkov, 2022; Li et al., 2023), these probes are simple linear classifiers trained on frozen hidden-state activations to identify MGT. We consider two variants: a **Layer-Averaged Linear Probe (LLP)**, which trains one linear probe per layer and averages predictions (Nordby et al., 2026); and a **Concatenated-Layer Linear Probe (CLP)**, which trains a single linear probe on the concatenation of hidden states.

Preliminaries Let $x_i = (x_1, \dots, x_{T_i})$ denote an input sequence of length T_i . An LLM processes x_i through L transformer layers and produces a hidden state $\mathbf{h}_{i,t}^{(\ell)} \in \mathbb{R}^d$ for each token position $t \in \{1, \dots, T_i\}$ and layer $\ell \in \{1, \dots, L\}$, where d is the hidden-state dimension. Given a binary MGT dataset, we extract the last-token hidden state from each layer, yielding the probing dataset $\mathcal{D}_{\text{probe}} = \{(\{\mathbf{h}_i^{(\ell)}\}_{\ell=1}^L, y_i)\}_{i=1}^N$, where $y_i \in \{0, 1\}$ indicates whether an instance is machine-generated ($y_i = 1$).

As Section 3 shows that MGT and HWT representations are linearly separable in low-dimensional spaces, we train our MGT probes on PCA-reduced activations. Specifically, for each layer, we fit PCA on the training activations and project them onto the top 100 principal components. With Llama-3-8B (Grattafiori et al., 2024) as the base model, this reduces the representation dimensionality to $100/4096 \approx 2.4\%$ of the original space.

We ablate this choice in Appendix D. Probe performance with and without PCA is nearly identical, indicating that PCA does not drive linear separability.

Layer-Averaged Linear Probe (LLP) We train an independent logistic regression classifier $f(\cdot)$ for each layer ℓ :

$$\min_{\mathbf{w}^{(\ell)}} \frac{1}{N} \sum_{i=1}^N \mathcal{L}(y_i, f(\mathbf{h}_i^{(\ell)}; \mathbf{w}^{(\ell)})) + \lambda \|\mathbf{w}^{(\ell)}\|_2^2. \quad (1)$$

where λ controls regularization and $\mathcal{L}$ is the binary cross-entropy loss.

After training, we normalize the weight vector $\mathbf{v}^{(\ell)} = \frac{\mathbf{w}^{(\ell)}}{\|\mathbf{w}^{(\ell)}\|_2}$, which defines the **probing vector**

for layer ℓ . Intuitively, $\mathbf{v}^{(\ell)}$ captures the linear direction in activation space that separates MGT from HWT (Gurnee and Tegmark, 2024).

At inference time, we pass a test instance x_{test} through the LLM, extract its last-token hidden state $\mathbf{h}_{\text{test}}^{(\ell)}$ at each layer, and project it onto the corresponding probing direction: $s_{\text{test}}^{(\ell)} = \mathbf{h}_{\text{test}}^{(\ell)\top} \mathbf{v}^{(\ell)}$ (Tigges et al., 2024). This projection measures how strongly the hidden state points in the MGT direction identified by the probe. We then compute the final LLP score by averaging the layer-wise projection scores: $s_{\text{LLP}}(x_{\text{test}}) = \frac{1}{L} \sum_{\ell=1}^L s_{\text{test}}^{(\ell)}$. Higher scores indicate stronger alignment with the MGT direction, and vice versa.

Concatenated-Layer Linear Probe (CLP) We also train a single linear probe on the concatenation of hidden states across layers. For each input x_i , we construct $\mathbf{h}_i^{\text{concat}} = [\mathbf{h}_i^{(1)}; \dots; \mathbf{h}_i^{(L)}] \in \mathbb{R}^{L \times 100}$, which we use to train the same classifier as in Equation 1 and perform inference analogously. Compared to LLP, CLP learns a single MGT probe vector that *jointly* identifies the most informative layers and the within-layer directions that encode MGT signals.

5 Experiments

5.1 Experimental Setup

We provide full details on the benchmarks, detectors, and implementations in Appendix B.

Benchmarks We evaluate detectors on 4 benchmarks, each comprising 4 subsets that capture different dimensions of MGT. We use DetectRL (Wu et al., 2024) to study *domains* (e.g., Reddit), Multi-Social (Macko et al., 2025) to analyze *languages* (e.g., Chinese), RAID (Dugan et al., 2024) to different *generators* (e.g., GPT4), and TSM (Quaremba et al., 2026) to examine *generation tasks* (e.g., summarization). For each of the 16 subsets, we randomly sample 1,500 training instances and 500 test instances, balanced across labels. Within each benchmark, the sampled data vary along additional dimensions (e.g., domains include multiple generators and adversarial attacks), resulting in more realistic and challenging detection settings.

Baselines We include 16 competitive MGT detectors. For zero-shot detectors we include *Log-Likelihood* (Solaiman et al., 2019), *LLR* (Su et al., 2023), *Rank* (Gehrmann et al., 2019), *GEC-Score* (Wu et al., 2025b), *Revise* (Zhu et al., 2023a),

Model	DetectRL (Wu et al., 2024)				MultiSocial (Macko et al., 2025)				RAID (Dugan et al., 2024)				TSM (Quaremba et al., 2026)			
	ArXiv	Reddit	Yelp	News	en	de	ru	zh	Cohere	GPT4	Llama	Mistral	FP	PE	SUM	TST
Zero-shot																
Likelihood	0.6534	0.8620	0.8852	0.5793	0.7902	0.6935	0.6720	0.5848	0.7025	0.6937	0.7867	0.7339	0.5214	0.4344	0.5045	0.5096
LLR	0.7317	0.8226	0.8660	0.5965	0.7759	0.7130	0.6333	0.6500	0.7577	0.7212	0.8406	0.8091	0.5025	0.4130	0.5147	0.5107
Rank	0.5727	0.3702	0.3016	0.6061	0.2141	0.2942	0.3506	0.3846	0.3768	0.3217	0.2223	0.2600	0.5203	0.5561	0.4634	0.4886
GECScore	0.7848	0.5631	0.6283	0.7065	0.8057	0.6714	0.6133	0.4374	0.6655	0.4620	0.5497	0.6642	0.5459	0.5168	0.6678	0.5445
Revise	0.8527	0.7535	0.7917	0.7710	0.7740	0.6409	0.6835	0.6377	0.7364	0.6230	0.6676	0.7536	0.5726	0.5688	0.6261	0.5760
RAIDAR	0.8566	0.5395	0.7158	0.8490	0.8131	0.7943	0.6446	0.9153	0.8454	0.7234	0.7682	0.8444	0.6883	0.6706	0.7377	0.6729
FastDetectGPT	0.8563	0.6969	0.8304	0.6842	0.7850	0.7185	0.6500	0.6251	0.8922	0.7441	0.8221	0.7752	0.5060	0.4347	0.6082	0.4963
Binoculars	0.9393	0.9423	0.9661	0.8682	0.8082	0.7473	0.7118	0.7428	0.9312	0.8960	0.9490	0.9207	0.5668	0.4930	0.6294	0.5429
Supervised																
EditLens	0.6732	0.4702	0.5050	0.3088	0.6769	0.7164	0.7510	0.6432	0.6532	0.5239	0.5835	0.5707	0.5840	0.6991	0.6092	0.6313
ID	0.7251	0.5616	0.6996	0.7681	0.5487	0.4998	0.5793	0.4900	0.5383	0.4832	0.5075	0.5280	0.5513	0.4715	0.5916	0.5312
OpenAI-RoBERTa	0.8357	0.7954	0.8101	0.7889	0.4683	0.3427	0.6985	0.4464	0.6855	0.5972	0.7070	0.7111	0.4483	0.5074	0.4309	0.5633
RADAR	0.9876	0.8733	0.9572	0.9976	0.6904	0.4682	0.3317	0.3392	0.8531	0.8689	0.8883	0.8896	0.5736	0.6302	0.4725	0.6192
RepreGuard	0.8960	0.9700	0.9780	0.9580	0.7060	0.6920	0.6400	0.7840	0.6020	0.7820	0.8920	0.8140	0.5040	0.5300	0.6040	0.4980
BiScope	0.9923	0.9820	0.9838	0.9925	0.8789	0.8014	0.7779	0.7904	0.9203	0.8791	0.9715	0.9481	0.7513	0.6011	0.7969	0.7056
TextFluoroscropy	0.9948	0.9983	0.9951	0.9980	0.9313	0.9015	0.8810	0.9032	0.8832	0.9546	0.9692	0.9492	0.7058	0.3723	0.2781	0.4089
RoBERTa	1.0000	0.9996	0.9994	1.0000	0.9507	0.8649	0.7762	0.8471	0.9404	0.9669	0.9770	0.9814	0.9361	0.7539	0.9333	0.7924
MGT Probes																
LLP	1.0000	1.0000	1.0000	1.0000	0.9565	0.9337	0.9212	0.9443	0.9714	0.9925	0.9946	0.9928	0.9707	0.9424	0.9783	0.8960
Δ vs BL	+0.00	+0.04	+0.06	+0.00	+0.58	+3.22	+4.02	+2.90	+3.09	+2.55	+1.76	+1.14	+3.46	+18.85	+4.50	+10.35
CLP	1.0000	0.9999	0.9998	1.0000	0.9511	0.9196	0.9056	0.9510	0.9652	0.9909	0.9949	0.9872	0.9725	0.9196	0.9697	0.8905
Δ vs BL	+0.00	+0.04	+0.04	+0.00	+0.04	+1.81	+2.46	+3.57	+2.48	+2.39	+1.78	+0.58	+3.64	+16.57	+3.64	+9.80

Table 1: In-domain detection AUC scores across four benchmarks, comparing 16 baseline detectors (zero-shot and supervised) with MGT probes. **Bold** indicates the best score per column, **underline** the second-best score, and “Δ vs. BL” the gain over the strongest baseline. LLP = Layer-Averaged Linear Probes; CLP = Concatenated-Layer Probes. For TSM, FP = First Paragraph, PE = Paragraph Extension, SUM = Summarization, and TST = Text Style Transfer. **Takeaway:** oth LLP and CLP consistently outperform all baselines by +0.04–18.85 AUC.

Raidar (Mao et al., 2024), *FastDetectGPT* (Bao et al., 2023), and *Binoculars* (Hans et al., 2024).

For supervised detectors we select *EditLens* (Thai et al., 2026), *ID* (Tulchinskii et al., 2023), *OpenAI-RoBERTa* (Solaiman et al., 2019), *RADAR* (Hu et al., 2023), *RepreGuard* (Chen et al., 2025), *BiScope* (Guo et al., 2024), *TextFluoroscropy* (Yu et al., 2024), and fully fine-tune RoBERTa-base (Liu et al., 2019).

Models and Metrics We implement probes with Llama-3-8B (Grattafiori et al., 2024) as the base model³ and sklearn for the logistic regression with a $L2$ penalty of $C=1$. Following prior work (Bao et al., 2023; Hans et al., 2024) we report AUC as the evaluation metric.

5.2 In-domain (ID) Detection

Our findings in Section 3 suggest that simple linear probes may provide effective detection, we therefore begin by asking **RQ1:** *How do MGT probes compare to MGT detectors in-domain?*

MGT probes consistently outperform baselines across 16 in-domain settings by 0.04–18.85 AUC. Table 1 reports AUC results across 16 ID settings, comparing 16 detectors with our MGT probes. Both variants consistently outperform baselines, achieving gains of up to 18.85 AUC relative

to the strongest detector. The near-identical performance of CLP and LLP suggests that a single probe trained on concatenated layer representations is sufficient for ID detection.

On DetectRL, baseline detectors exhibit near-saturated performance despite the domain subsets containing text generated by multiple LLMs and subjected to diverse adversarial attacks (e.g., paraphrasing). As the benchmark provides little headroom for improvement, probes achieve only relatively limited gains.⁴ On MultiSocial, probes achieve the largest improvements over baselines on non-English languages (+1.81–4.02 AUC), suggesting that the learned probing directions are language-agnostic, for which we find support in OOD (§5.3) and probing vector experiments (§5.4). Across generators on RAID, probes show additional robust gains over already strong baselines (+0.58–3.09 AUC). TSM presents the most challenging detection setting, where even supervised detectors struggle with mixed human–machine text (PE) and shorter texts containing only minor stylistic edits (TST). Nevertheless, MGT probes maintain strong performance, achieving the largest gains over baselines (+3.46–18.85 AUC).

³We ablate model architectures and sizes in Appendix D.

⁴We also experimented with domains from RAID and M4GT, observing similar saturation effects.